

Project 1: The Sentinel (FIM)

1. Overview

A Python-based **File Integrity Monitor (FIM)** designed to ensure data integrity by detecting unauthorized changes in a target directory (`/vault`).

2. How it Works (The Logic)

The script uses the **SHA-256** hashing algorithm. Since every file has a unique hash based on its content, the script can detect even a single bit of change.

- **Baseline:** On startup, it captures the "known good" state of all files.
- **Polling:** Every 2 seconds, it re-scans and compares the new hashes against the baseline.
- **Alerting:** It triggers specific flags for **New**, **Modified**, or **Deleted** files.

3. Key Commands Used

Task	Command
Check Hash Manually	<code>sha256sum vault/test.txt</code>
Run Sentinel	<code>python3 sentinel.py</code>
Simulate Attack	<code>echo "malware" >> vault/test.txt</code>

```
(kali㉿kali)-[~/SentinelProject]
$ touch ~/SentinelProject/vault/hacker.txt

(kali㉿kali)-[~/SentinelProject]
$ echo "malware code" >> ~/SentinelProject/vault/test.txt

(kali㉿kali)-[~/SentinelProject]
$ echo "malware code" >> ~/SentinelProject/vault/hacker.txt

(kali㉿kali)-[~/SentinelProject]
$
```

sentinel.py vault

```
(kali㉿kali)-[~/SentinelProject]
$ sudo nano sentinel.py

(kali㉿kali)-[~/SentinelProject]
$ cd vault

(kali㉿kali)-[~/SentinelProject/vault]
$ echo "hello world" > ~/SentinelProject/vault/test.txt

(kali㉿kali)-[~/SentinelProject/vault]
$ sha256sum ~/SentinelProject/vault/test.txt
a948904f2f0f479b8f8197694b30184b0d2ed1c1cd2a1ec0fb85d299a192a447 /home/kali/
SentinelProject/vault/test.txt

(kali㉿kali)-[~/SentinelProject/vault]
$ python3 sentinel.py
python3: can't open file '/home/kali/SentinelProject/vault/sentinel.py': [Err
no 2] No such file or directory

(kali㉿kali)-[~/SentinelProject/vault]
$ cd ../

(kali㉿kali)-[~/SentinelProject]
$ python3 sentinel.py
🛡 SENTINEL ACTIVE: Monitoring /home/kali/SentinelProject/vault
📄 NEW FILE: /home/kali/SentinelProject/vault/hacker.txt
⚠ MODIFIED: /home/kali/SentinelProject/vault/test.txt
⚠ MODIFIED: /home/kali/SentinelProject/vault/hacker.txt
█
```